Miguel González

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Research interests

Complex and algebraic geometry, moduli spaces of geometric structures on Riemann surfaces, Higgs bundles, mirror symmetry, Langlands duality.

Experience

Postdoctoral Researcher 2026-
Institute of Science and Technology Austria (ISTA)
Research group of Tamás Hausel

PhD Research - *la Caixa* foundation InPhINIT Retaining Fellowship 2023-2026
Instituto de Ciencias Matemáticas (ICMAT) *Project: Classification of very stable Higgs bundles*
Advisers: Oscar García-Prada and Tamás Hausel *Grant LCF/BQ/DR23/12000030*

Research Grant Sept. 2022 - Jun. 2023
Instituto de Ciencias Matemáticas (ICMAT) *Project: Cyclic Higgs bundles and Vinberg theory*
Supervised by Oscar García-Prada *Grant JAEICU-21-ICMAT-01, Top 1 application score*

Research Intern Jul. 2022 - Aug. 2022
Hausel Group (Geometry and its Interfaces), Institute Of Science And Technology Austria (ISTA) *Project: Very stable $U(p,q)$-Higgs bundles*
Supervised by Tamás Hausel

Collaboration grant Nov. 2021 - Jul. 2022
Department of Mathematics, Universidad Autónoma de Madrid *Project: Theoretical and computational research about*
Supervised by Ernesto Girondo Sirvent *properties of fundamental domains of Fuchsian groups*

Introduction to Mathematical Research Grant Severo Ochoa Summer 2021
Instituto de Ciencias Matemáticas (ICMAT) *Project: Introduction to Moduli Spaces and Higgs Bundles*
Supervised by Oscar García-Prada

Education

PhD Student in Mathematics Research 2023-2026
Universidad Complutense de Madrid (UCM) - Instituto de Ciencias Matemáticas (ICMAT) *"International PhD" distinction*
Advisers: Oscar García-Prada and Tamás Hausel

Masters Degree in Mathematics 2022 – 2023
Universidad Autónoma de Madrid (UAM) *Average grade: 9.94 / 10*
Dissertation: Cyclic Higgs bundles and Vinberg theory *5 courses with honours*

Double Degree in Mathematics and Computer Science 2017 – 2022
Universidad Autónoma de Madrid (UAM) *Average grade: 9.82 / 10*
44 courses with honours

Preprints

Very stable parabolic G-Higgs bundles and affine flag varieties June 2026
arXiv:2606.16880

Very stable regular G-Higgs bundles March 2025
arXiv:2503.01289

Publications

Cyclic Higgs bundles and the Toledo invariant — March 2026
With Oscar García-Prada
Transformation Groups, Vol. 31, 1505–1542
doi:10.1007/s00031-026-09962-2

Hitchin map on even very stable upward flows — April 2024
With Tamás Hausel
International Journal of Mathematics, Vol. 35, No. 9
doi:10.1142/S0129167X2441009X

Talks

Classification of very stable Higgs bundles — September 2026
Geometry conference in honour of Nigel Hitchin on the occasion of his 80th birthday
Instituto de Ciencias Matemáticas (ICMAT)

Very stable parabolic G-Higgs bundles and affine flag varieties — August 2026
Workshop: Unifying Geometric Perspectives on Hitchin's Moduli Space
Bernoulli Center, École polytechnique fédérale de Lausanne (EPFL)

Wobbly Higgs bundles and the Toledo invariant — April 2026
i-Link geometry workshop
Institute of Science and Technology Austria (ISTA)

Wobbly parabolic G-Higgs bundles and affine flag varieties — February 2026
Geometric Structures and Stability Thematic Program
International Centre for Theoretical Sciences (ICTS-TIFR)

Very stable G-Higgs bundles and Hecke modifications — November 2025
MIST Conference in Mathematical Physics
The Chinese University of Hong Kong (CUHK)

The structure of symmetry: an introduction to root systems — September 2025
ICMAT-UAM Junior Colloquium
Instituto de Ciencias Matemáticas (ICMAT)

A criterion for wobbliness of parabolic G-Higgs bundles — August 2025
Seminar talk
Chern Institute of Mathematics, Nankai University (CIM)

Compact components in relative character varieties for real lie groups of Hermitian type — August 2025
Geometry Workshop of the Hitchin–Ngô Laboratory
Feishu (Shanghai)

$\mathbb{C}^\times$-actions, lagrangian submanifolds and very stable bundles — June 2025
Workshop: Geometry and Topology for the Future III
Universidad Complutense de Madrid (UCM)

Very stable regular G-Higgs bundles — June 2025
Hitchin–Ngô Laboratory Special Activity
Instituto de Ciencias Matemáticas (ICMAT)

Very stable regular G-Higgs bundles — May 2025
Séminaire d'Analyse et Géométrie
Institut de Mathématiques de Jussieu–Paris Rive Gauche (IMJ-PRG)

Very stable regular G-Higgs bundles — May 2025
Conference: Stratifications of Higgs bundle moduli spaces and related topics
Universidade de Santiago de Compostela (USC)

Hitchin map on minuscule Lagrangians — February 2025
Bonn–Vienna block seminar: Equivariant cohomology, stable envelopes and big algebras
University of Bonn

The decomposition theorem I: finite fields — January 2025
Reading seminar on perverse sheaves
Universidad Complutense de Madrid (UCM)

Very stable G-Higgs bundles with regular nilpotent Higgs field — November 2024
RTGF (Spanish Geometry and Physics Thematic Network) meeting
Instituto de Ciencias Matemáticas (ICMAT)

The $\mathbb{C}^\times$-action on the moduli space of Higgs bundles — November 2024
"Doc in progress" seminar
Università degli Studi di Trento (UniTrento)

The classification of extended 2-TQFTs — May 2024
Reading seminar on ∞-categories
Universidad Complutense de Madrid (UCM)

Mixed Hodge theory — April 2024
Arbeitsgemeinschaft: Geometry and Representation Theory around the P=W Conjecture
Matematisches Forschungsinstitut Oberwolfach (MFO)

Cyclic Higgs bundles and the Toledo invariant — January 2024
Hausel group seminar
Institute of Science and Technology Austria (ISTA)

Introduction to Hodge theory in complex algebraic geometry — January 2024
PhD seminar
Universidad complutense de Madrid (UCM)

The Hitchin Fibration and Spectral Curves — February 2023
Hitchin-Ngô Lab Working Group
Instituto de Ciencias Matemáticas (ICMAT)

Attended conferences

Geometry conference in honour of Nigel Hitchin on the occasion of his 80th birthday — 21-25 September 2026
Instituto de Ciencias Matemáticas (ICMAT)

Trends in Differential and Complex Geometry: a celebration of Nigel Hitchin's 80th birthday — 31 August-4 September 2026
Mathematical Institute (University of Oxford)

Workshop: Unifying Geometric Perspectives on Hitchin's Moduli Space — 10-14 August 2026
Bernoulli Center, École polytechnique fédérale de Lausanne (EPFL)

2nd MaPhyAG Workshop: Algebraic Geometry and Mathematical Physics — 29 June - 3 July 2026
Universidad Complutense de Madrid (UCM)

i-Link geometry workshop — 7-10 April 2026
Institute of Science and Technology Austria (ISTA)

Geometric Structures and Stability Thematic Program — 16-27 February 2026
International Centre for Theoretical Sciences (ICTS-TIFR)

Hong Kong Laureate Forum 2025 — 5-8 November 2025
Hong Kong Science Park

MIST Conference in Mathematical Physics — 31 October-4 November 2025
The Chinese University of Hong Kong (CUHK)

Meeting of the Spanish Network of Geometry and Physics 2025 — 15-17 September 2025
Universitat de Barcelona (UB)

Geometry Workshop of the Hitchin–Ngô Laboratory — 7-9 August 2025
Feishu (Shanghai)

Feishu China Mathematics Symposium — 4-6 August 2025
Feishu (Shanghai)

Higher rank geometric structures, Higgs bundles and physics — 19-23 May 2025
Institut Henri Poincaré (IHP)

Stratifications of Higgs bundle moduli spaces and related topics — 5-9 May 2025
Universidade de Santiago de Compostela (USC)

Geometry at Large, an ICMAT–IMSA collaboration — 21-25 April 2025
Instituto de Ciencias Matemáticas (ICMAT)

RTGF (Spanish Geometry and Physics Thematic Network) meeting — 26-29 November 2024
Instituto de Ciencias Matemáticas (ICMAT)

1st MaPhyAG Workshop: Integrability, Geometry and QFT — 7-11 October 2024
Universidad Complutense de Madrid (UCM)

Clay Research Workshop: The P=W Conjecture in Non Abelian Hodge Theory — 30 September-4 October 2024
Mathematical Institute (University of Oxford)

2nd Simons Math Summer Workshop: Moduli — 1-19 July 2024
Simons Center for Geometry and Physics (SCGP)

Arbeitsgemeinschaft: Geometry and Representation Theory around the P=W Conjecture — 1-5 April 2024
Matematisches Forschungsinstitut Oberwolfach (MFO)

The Hitchin system, Langlands duality and mirror symmetry — 24-28 April 2023
Instituto de Ciencias Matemáticas (ICMAT)

Moduli spaces and geometric structures — 12-16 September 2022
Conference in honour of Oscar García-Prada on the occasion of his 60th birthday
Instituto de Ciencias Matemáticas (ICMAT)

ICMAT - JAE School of Mathematics — July 2021
Instituto de Ciencias Matemáticas (ICMAT)

Poster presentations

Very stable G-Higgs bundles — November 2025
Hong Kong Laureate Forum 2025
Hong Kong Science Park

Very stable $U(p,q)$-Higgs bundles — August 2022
Summer ISTerns poster session
Institute of Science and Technology Austria (ISTA)

Teaching

Linear algebra (Álgebra lineal) — Year 2025-2026, 1st semester
Linear algebra problem sessions for first year undergraduate students in mathematics.
Universidad Complutense de Madrid (UCM)

Pequeño Instituto de Matemáticas (PIM) / Little mathematics institute — 2023-2026
Advanced mathematics problem-solving classes for high school students. https://www.icmat.es/PIM/
Instituto de Ciencias Matemáticas (ICMAT)

Research visits

Institute of Science and Technology Austria (ISTA) — 15 March - 17 May 2026
Hausel Group (Geometry and its Interfaces)

Institut Henri Poincaré, Paris — 12-23 May 2025
Higher rank geometric structures research program

Faculty of Mathematics, University of Duisburg-Essen — 7-11 April 2025
Heinloth Group

Institute of Science and Technology Austria (ISTA) — 23 February - 22 March 2025
Hausel Group (Geometry and its Interfaces)

Institute of Science and Technology Austria (ISTA) — 15-26 January 2024
Hausel Group (Geometry and its Interfaces)

Institute of Science and Technology Austria (ISTA) — 19-28 January 2023
Hausel Group (Geometry and its Interfaces)

Awards and prizes

Master's degree Extraordinary Award (best academic record) — 2024
Awarded by Universidad Autónoma de Madrid (UAM)

UAM Young Talent Award — 2022
Awarded by Universidad Autónoma de Madrid (UAM)

Best undergraduate academic record — 2022
Awarded by Universidad Autónoma de Madrid (UAM)

Excellence Grant — 2018, 2019, 2020, 2021
Awarded by the Comunidad de Madrid

Best academic record of first year undergraduate students — 2018
Awarded by Universidad Autónoma de Madrid (UAM)

Concurso de Primavera de Matemáticas (Mathematics contest) 2017
Winner (Top 20)

Extraordinary Award of Bachillerato 2017
Awarded by the Comunidad de Madrid to Top <1% high secondary school students

Extraordinary Award of Secondary Education 2015
Awarded by the Comunidad de Madrid to Top <1% basic secondary school students

Miscellaneous publications

Report for Arbeitsgemeinschaft: Geometry and Representation Theory around the P=W Conjecture
Editor and contributor with the extended abstract for my talk on Mixed Hodge theory.
doi:10.14760/OWR-2024-16

Undergraduate course notes
LaTeX notes for every course I took during my undergraduate mathematics studies.
Available at miguelgg.com

Declined grants (incompatibilities)

PhD studies grant 2023
FPU Grant - Spanish Ministry of Universities
Funding for a 4 year PhD research and teaching contract

PhD studies grant 2023
ICMAT Severo Ochoa FPI Grant - Spanish Ministry of Science and Innovation
Funding for a 4 year PhD research contract

Masters studies grant 2022-2023
Department of Mathematics, Universidad Autónoma de Madrid (UAM) *Top 1 application score*
10000€ for masters studies and research

Skills

Spanish: Native level
English: Fluent level
French: Intermediate level
Japanese: Intermediate level (comprehension), basic level (production)
LaTeX knowledge
Computer skills: Python, C, Java, GitHub, Databases (SQL), Machine Learning and AI (Master rank in kaggle.com), among others.